

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech **Branch :Computer Engineering/Computer Science Engineering(AIDS) and Allied**

Subject Code & Name: BTCOC304 & Computer Architecture & Organization **Semester :III**

Max. Marks: 60

Date: 12/02/2025

Duration: 3 H

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Which component acts as the "brain" of a computer and is responsible for executing instructions?	1	1				
	<table border="1"> <tr> <td>a. Memory</td> <td>b. CPU</td> <td>c. ALU</td> <td>d. Hard Disk</td> </tr> </table>	a. Memory	b. CPU	c. ALU	d. Hard Disk		
a. Memory	b. CPU	c. ALU	d. Hard Disk				
	The part of the CPU responsible for performing arithmetic and logical operations is	1	1				
	<table border="1"> <tr> <td>a. Control Unit</td> <td>b. Arithmetic Logic Unit (ALU)</td> <td>c. Register</td> <td>d. Cache Memory</td> </tr> </table>	a. Control Unit	b. Arithmetic Logic Unit (ALU)	c. Register	d. Cache Memory		
a. Control Unit	b. Arithmetic Logic Unit (ALU)	c. Register	d. Cache Memory				
3	Which of the following is NOT a type of addressing mode?	2	1				
	<table border="1"> <tr> <td>a. Immediate</td> <td>b. Direct</td> <td>c. Indirect</td> <td>d. Inline</td> </tr> </table>	a. Immediate	b. Direct	c. Indirect	d. Inline		
a. Immediate	b. Direct	c. Indirect	d. Inline				
4	The process of fetching, decoding, and executing an instruction is called	2	1				
	<table border="1"> <tr> <td>a. Instruction Cycle</td> <td>b. Machine Cycle</td> <td>c. Fetch Cycle</td> <td>d. Execute Cycle</td> </tr> </table>	a. Instruction Cycle	b. Machine Cycle	c. Fetch Cycle	d. Execute Cycle		
a. Instruction Cycle	b. Machine Cycle	c. Fetch Cycle	d. Execute Cycle				
5	The binary equivalent of the decimal number 13 is:	2	1				
	<table border="1"> <tr> <td>a. 1011</td> <td>b. 1101</td> <td>c. 0010</td> <td>d. 1111</td> </tr> </table>	a. 1011	b. 1101	c. 0010	d. 1111		
a. 1011	b. 1101	c. 0010	d. 1111				
6	Which of the following is NOT a positional number system?	2	1				
	<table border="1"> <tr> <td>a. Binary</td> <td>b. Decimal</td> <td>c. Roman</td> <td>d. Octal</td> </tr> </table>	a. Binary	b. Decimal	c. Roman	d. Octal		
a. Binary	b. Decimal	c. Roman	d. Octal				
7	In binary arithmetic, what is the 2's complement of 1010?		1				

	a. 0101	b. 0110	c. 1010	d. 0110		
8	In IEEE 754 single-precision format, how many bits are allocated for the exponent?				2	1
	a. 8	b. 10	c. 23	d. 32		
9	The data transfer method where CPU is involved in the transfer is called:				3	1
	a. DMA	b. Programmed I/O	c. Interrupt-driven I/O	d. Cache I/O		
10	Which of the following is a secondary storage device?				3	1
	a. Cache	b. SSD	c. DRAM	d. Registers		
11	Microprogramming is a technique used to design				4	1
	a. Memory Hierarchy	b. Control Units	c. ALU	d. Registers		
12	The Control Unit generates which of the following signals?				5	1
	a. Control Signals	b. Arithmetic Signals	c. Data Transfer Signals	d. Machine Language Signals		
Q. 2	Solve the following.					12
A)	Explain briefly main structural components of computer				1	6
B)	Define concept Computer organization and Computer Architecture, Structure and Function				1	6
Q.3	Solve the following.					12
A)	List Addressing modes with example				2	6
B)	Differentiate RISC Vs CISC Architecture				2	6
Q. 4	Solve Any Two of the following.					12
A)	Calculate 1's and 2's Complement of following a) 12 b) 18 c) 27 d)9				3	6
B)	Perform Multiplication Using Booths Algorithm i) 12 x 13 ii) -14 x 15				3	6
C)	Write Steps of Binary Division restoring algorithm with example				3	6
Q.5	Solve Any Two of the following.					12

A)	Explain Cache Memory with Property "Locality of reference" and replacement algorithm.	4	6
B)	What is mapping function? explain Direct mapping method with neat diagram.	4	6
C)	Write Short Note on Magnetic Disk, Magnetic Tape and Optical Memory.	4	6
Q.6	Solve Any Two of the following.		12
A)	Explain Microprogrammed control Unit with neat diagram.	5	6
B)	Write a short note on I/O channels, Direct Memory Access (DMA) and I/O Module.	5	6
C)	List the phases of Microinstruction Execution.	5	6
*** End ***			

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